

Zain Rizvi

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EDUCATION

University of Ottawa

Honours BSc Computer Science, Co-op

Ottawa, ON

Sept. 2023 – Dec. 2027

PROJECTS

CacheLayer | *Python, FastAPI, sentence-transformers, SQLite, Docker, Railway*

- [cachelayer.up.railway.app](#) | [Paper](#) | [System Design](#) | [GitHub](#)
- Built and deployed a multi-tenant semantic caching proxy for LLM APIs supporting OpenAI and Anthropic, intercepting paraphrased queries using sentence embeddings and cosine similarity with per-user API key authentication and isolated cache stores
- Achieved 93.3% cache hit rate at a 0.80 cosine similarity threshold across 15 paraphrased query pairs, returning cached responses in 8ms versus 1,800ms for live API calls; validated with a 10-test pytest suite and GitHub Actions CI
- Containerized with Docker for production deployment; automated image builds and pushes to GitHub Container Registry via GitHub Actions CI/CD pipeline with layer caching
- Empirically characterized the threshold-hit rate tradeoff across six similarity values and three query domains; published findings as a [technical paper](#)

SalesRadar | *React, Node.js, Express, IBM Watson NLU, Salesforce API, NewsAPI, Vercel, Railway*

- [sales-radar-seven.vercel.app](#) | [Demo](#) | [GitHub](#)
- Built an account intelligence platform running real-time news through IBM Watson NLU to extract buying signals across five categories, producing a 0–100 purchase intent score with priority tier and per-signal evidence
- Integrated Salesforce CRM via OAuth 2.0 REST API, enabling one-click export of scored accounts as Leads with the full Watson intelligence report attached

NFL Contract Value Analyzer | *Python, XGBoost, scikit-learn, K-Means, pandas, Streamlit, pytest, Plotly*

- [nfl-contract-analyzer.streamlit.app](#) | [Demo](#) | [GitHub](#)
- Trained separate XGBoost regression models per position to predict fair market APY for 264 active NFL players, achieving R^2 of 0.96 for QBs and MAE under \$1M for WRs and TEs
- Joined nflverse stats with Over The Cap contract data across 50,000+ records and applied K-Means clustering to discover 4 unsupervised player archetypes per position; validated pipeline with pytest unit tests covering data ingestion, feature engineering, and model output assertions

Court Vision — NBA Analytics | *Python, PostgreSQL, Streamlit, Plotly, Supabase, Docker, Dagster, dbt*

- [nbaanalytics9.streamlit.app](#) | [Demo](#) | [GitHub](#)
- Built an end-to-end analytics pipeline ingesting 1,400+ player-season records via nba.api into a PostgreSQL star schema on Supabase, computing a composite Player Value Score across five performance dimensions
- Modelled PVS transformation logic across staging and mart layers using dbt; orchestrated ingestion jobs with Dagster and containerized the pipeline with Docker for reproducible deployments

EXPERIENCE

IT Analyst

May 2024 – Aug. 2024 | May 2025 – Aug. 2025

York Region (*The Regional Municipality of York*)

Newmarket, ON · Hybrid

- Triaged and investigated 500+ monthly IT incidents across 15+ service categories, identifying recurring patterns and coordinating with technicians to resolve root causes
- Built incident tracking dashboards in ServiceNow, Excel, and Power BI monitoring volume, priority, and resolution time across a team of 10+ technicians, with findings informing infrastructure and staffing decisions
- Wrote SQL queries and Python scripts to automate recurring IT reporting tasks, reducing manual effort and improving data consistency across operational workflows
- Standardized ticket taxonomy and documentation practices to improve data consistency and enable accurate security and operational reporting for management

Director of Academic Affairs

Sept. 2023 – May 2024

Computer Science Student Association (CSSA)

University of Ottawa

- Organized professional development events including a Nokia office tour for 90+ CS students and workshops on web development and machine learning reaching 50+ students per session
- Represented 900+ undergraduate CS students as the primary liaison to department administration on academic policy and course scheduling

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java, HTML/CSS, R

AI & ML: sentence-transformers, XGBoost, scikit-learn, IBM Watson NLU, pandas, K-Means, Plotly, Streamlit

Web & APIs: React, Node.js, Express, FastAPI, MongoDB, REST APIs, OAuth 2.0, Salesforce API

Tools: Git, Docker, AWS, PostgreSQL, Power BI, Excel, Dagster, dbt, pytest, GitHub Actions, Railway, Nginx, Agile